Ancient Machine 2 (Solution)

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Introduction

In this review, we only explain solutions which lead to full score solutions. However, there are other solutions which give good results but do not lead to full score solutions.

In the following, the indices of elements of arrays are zero-based (i.e. they start from 0), and "the value set in the memory area of the machine" is called "the state" of the machine.

$M = 1002$ (10 points)

We can determine the i-th character by $m = i + 3$. Concretely, we construct the sequences as follows.

$$(a_j, b_j) = \begin{cases} (j+1, j+1) & 0 \leq j \leq i-1 \\ (i+1, i+2) & j = i \\ (j, j) & j = i+1, i+2 \end{cases}$$

The i-th character is '`0`' if the final state is $i + 1$, and it is '`1`' if the final state is $i + 2$.

Since $N = 1000$, we determine each character one-by-one from the first one, and get a solution with $M = 1002$.

$M = 1001$ (10 points)

For the $|T|$ characters from the last one, it is possible to determine whether the string S coincides with the string T or not by $m = |T| + 1$.

For example, if $T =$`"101"`, we construct as follows.

- $a = [0, 2, 0, 2]$
- $b = [1, 1, 3, 1]$

The current state is i if the characters are the same until the $(i - 1)$-th character of T.

If the final state is $|T|$, the last characters coincide with T; otherwise, the last characters do not coincide with T.

By this method, we determine the characters from the last one, and get a solution with $M = 1001$.

$M = 502$ (37 points)

We use an $M = 1002$ solution as above for the first half of characters (500 characters), and use an $M = 1001$ solution as above for the last half of characters (500 characters). Combining them, we get a solution with $M = 502$.

$M = 114$ (97 points)

We can achieve $M = 114$ using linear algebra.

There are 1000 unknown quantities. We would like to have 1000 independent linear equations in 1000 variables. Since we know each quantity is either 0 or 1, we can consider equations mod 2.

Concretely, we proceed as follows.

We get the sum of S_i's (mod 2) for every i with $i \equiv q \pmod{p}$ by $m = 2p$. Here we choose the values of p and q by row reduction (or Gaussian elimination) so that the vectors for chosen p and q (for example, $[0, 1, 0, 0, 1, 0, 0, 1, 0, \dots]$ if $p = 3$ and $q = 1$) are linearly independent.

We can find 1000 linearly independent vectors in the range $p \leq 57$. Hence we get a solution with $M = 114$.

$M = 102$ (100 points)

Combining an $M = 114$ solution and an $M = 502$ solution, we achieve $M = 102$ as follows.

To get information on a particular character amounts to choose the vector $[0, 0, \dots, 0, 1, 0, \dots, 0]$.

Up to 102, we determine the first 100 characters and the last 101 characters; hence we get 201 vectors. Additionally, in the range $p \leq 51$, we get 799 vectors such that the $201 + 799 = 1000$ vectors are linearly independent. Hence we get a solution with $M = 102$.